EXPERIMENT NO. 9

MINI PROJECT: STUDENT MARKS PREDICTION USING LINEAR REGRESSION

Subject Code: AI23521

Subject Name: Build and Deploy for Machine Learning Application

Submitted by: S.satyajit

Register Number: 241501192

Department: Artificial Intelligence and Machine Learning

Academic Year: 2026–2027

---

1. AIM

To develop a Machine Learning model using the Linear Regression algorithm to predict student marks based on the number of study hours and evaluate the model's performance using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score.

2. ABSTRACT

Student performance prediction is an application of Machine Learning that estimates academic marks based on study hours. This mini-project uses the Linear Regression algorithm to identify the relationship between study hours and marks.

A sample dataset is created using Python and processed using the Pandas library. The dataset is divided into training and testing sets. A Linear Regression model is trained using the training data and evaluated using standard regression metrics. Graphs are generated to compare actual and predicted marks. The trained model is also used to predict marks for a new student based on the entered study hours.

The project is implemented using Google Colab and Python libraries.

3. OBJECTIVES

1. To create a dataset containing study hours and student marks.
2. To preprocess and prepare the dataset for Machine Learning.
3. To train a Linear Regression model.
4. To predict marks using the trained model.
5. To evaluate model performance using regression metrics.
6. To visualize the relationship between study hours and marks.
7. To demonstrate a simple Machine Learning prediction system.

4. SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

- Google Colab
- Python 3
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Web browser and internet connection

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Internet connection

5. ALGORITHM

1. Start the program.
2. Import the required Python libraries.
3. Create a sample dataset containing study hours and corresponding marks.
4. Check the dataset structure and missing values.
5. Select study hours as the input feature and marks as the target variable.
6. Divide the dataset into training and testing sets using an 80:20 ratio.
7. Initialize the Linear Regression model.
8. Train the model using the training dataset.
9. Predict marks for the test dataset.
10. Calculate MSE, RMSE, and R^2 score to evaluate the model.
11. Plot actual versus predicted marks.
12. Plot the regression line representing the relationship between study hours and marks.
13. Accept study hours for a new student and predict the marks.
14. Display the prediction result.

15. Stop the program.

6. METHODOLOGY

6.1 Dataset Creation

A sample dataset is created using Python and Pandas. It contains two columns: Study_Hours and Marks. The dataset consists of 20 sample records.

- **Study_Hours:** Number of hours spent studying.
- **Marks:** Corresponding marks obtained by the student.

The dataset is used to demonstrate the relationship between study hours and academic marks.

```
   Study_Hours  Marks
0          1.0     20
1          2.0     25
2          3.0     35
3          4.0     40
4          5.0     50
Dataset shape: (20, 2)
Missing values: 0
```

6.2 Data Preprocessing

The dataset is converted into a Pandas DataFrame. The data is checked for missing values, and the input and target variables are selected.

- Input feature (X): Study_Hours
- Target variable (y): Marks

6.3 Train-Test Split

The dataset is divided into training and testing sets using the train_test_split() function from Scikit-learn.

- Training data: 80% of the dataset (16 records).
- Testing data: 20% of the dataset (4 records).

The training data is used to train the model, while the testing data is used to evaluate its performance.

```
Training samples: 16
Testing samples: 4
Model training completed successfully!
```

6.4 Model Training

The Linear Regression algorithm is implemented using the LinearRegression() class from Scikit-learn.

The model learns the relationship between study hours and marks from the training data. Linear Regression estimates the target value using a best-fit straight line.

The general equation is:

y = mx + c

Where:

- y = Predicted marks
- x = Study hours
- m = Slope of the regression line
- c = Intercept

The model parameters are estimated during training.

6.5 Prediction

After training, the model predicts marks for the testing dataset. It can also predict marks for a new student by accepting the number of study hours as input.

The predicted result is displayed in the form of marks out of 100. The output is limited to the range of 0–100 for the new-student demonstration.

6.6 Model Evaluation

The model is evaluated using the following metrics:

Mean Squared Error (MSE): Measures the average squared difference between actual and predicted marks. Lower values generally indicate smaller errors.

Root Mean Squared Error (RMSE): Represents the square root of MSE and expresses prediction error in the same units as marks.

R^2 Score: Measures how well the model explains the variation in the target values. A value closer to 1 generally indicates a better fit.

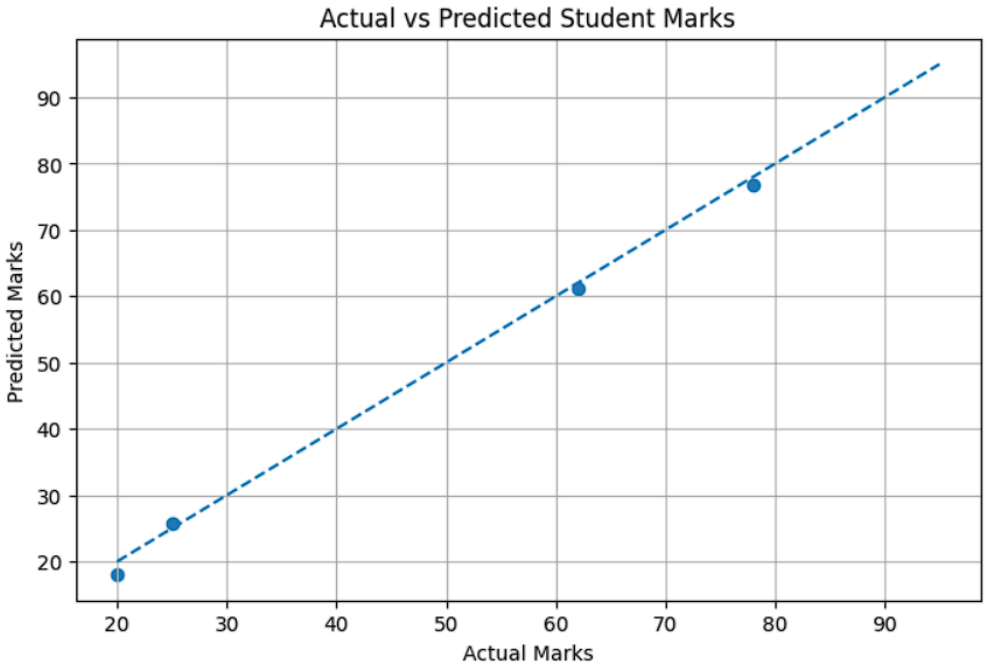
```
Mean Squared Error (MSE): 1.66
Root Mean Squared Error (RMSE): 1.29
R2 Score: 0.997
   Actual Marks  Predicted Marks
0            20            18.00
1            78            76.89
2            62            61.19
3            25            25.85
```

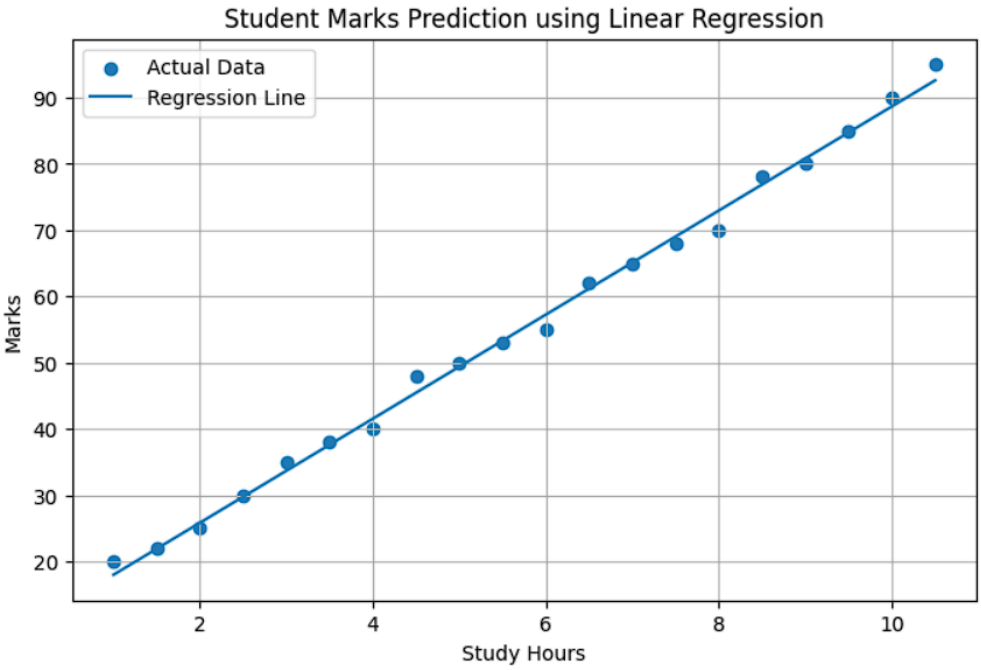
7. IMPLEMENTATION

The project is implemented in Google Colab using Python. The implementation consists of the following stages:

1. Importing libraries and creating the dataset.
2. Splitting the dataset into training and testing sets.
3. Training the Linear Regression model.
4. Evaluating model performance.
5. Visualizing actual and predicted marks.
6. Predicting marks for a new student.

The complete Python source code is provided in the accompanying Google Colab notebook named Student_Marks_Prediction.ipynb.






```
Enter study hours: 7
Study Hours: 7.0
Predicted Marks: 65.11 / 100
```

8. OUTPUT AND DISCUSSION

The program successfully creates a sample dataset and trains a Linear Regression model using study hours as the input feature and marks as the target variable.

The evaluation metrics indicate the model's performance on the test dataset. The actual-versus-predicted graph helps visualize prediction errors, while the regression line graph illustrates the relationship between study hours and marks.

The new-student prediction cell accepts study hours as input and displays the model's estimated marks.

The results are based on a small illustrative dataset created for educational purposes. Therefore, the predictions demonstrate the working principle of Linear Regression and should not be treated as reliable forecasts of actual student performance.

9. RESULT

The Student Marks Prediction mini-project was implemented successfully using the Linear Regression algorithm in Google Colab. The model was trained using a sample dataset containing study hours and marks. Predictions were generated for test data and a new student, and model performance was evaluated using MSE, RMSE, and R^2 score. Graphs were plotted to visualize actual and predicted marks and the regression line.

Thus, the implementation demonstrates the use of Linear Regression for predicting a numerical value from a given input feature.